ARRAY

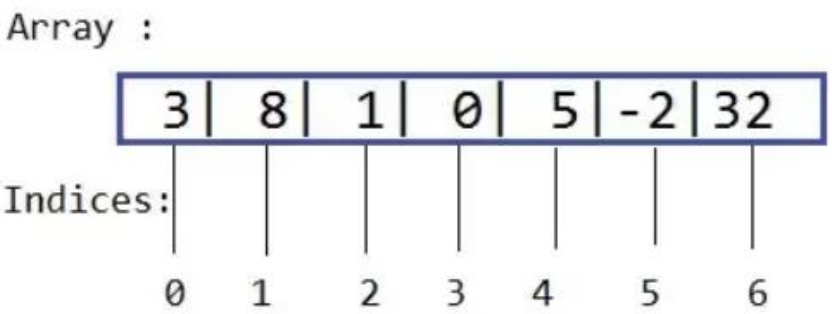
-- Second Lecture

Created By
ABDULLAH AL RAFI & AJDM Jakiur Rahman

WHAT IS ARRAY?

An array can be defined as a group of similar kinds of elements or data items that are stored together in contiguous memory spaces

Ex: int type array (int arr[4];) , character type array (char arr[4];).



ENOUGH OF POWERPOINT…

Let’s Dive Into Our IDE

(INTEGRATED DEVELOPMENT ENVIRONMENT)

Knowledge is of no value unless you put it into PRACTICE .

– Anton Chekhov.

```
// defining array:
int n = 5; //variable

int a[8]; //array
int b[n]; //array

char c[7]; //array
char d[n]; //array

// give elements to array:
int e[] = {3, 2, 5, 1}; //giving 4 elements
cout << e[4] << endl; //index 4 is
cout << e[0] << " " << e[2] << endl; //variable

char h[] = {'a', 'e', 'i', 'o', 'u'};
cout << h[0] << ' ' << h[2] << ' ' << h[5] << endl;

// take input in array:
for (int i = 0; i < n; i++)
{
    cin >> b[i];
} //same in char array

for (int i = 0; i < n; i++)
{
    cout << b[i];
}
cout << endl;

change a value:

int f[10] = {0};
for (int i = 0; i < 10; i++)
{
    cout << f[i] << ' ';
}

// globally declared array auto assigned 0 as values;

// sorting + maximum && minimum;
int arr[] = {5,4,3,5,6,3,5,7,8};
sort(arr,arr+n);
reverse(arr,arr+n);
```

Array Mane Ki Vai ????

Matrix !!!!!
Asolei ????

$$
\begin{array}{c}
 \\ 1 \\ 2 \\ 3 \\ \vdots \\ m
\end{array}
\begin{array}{c}
\begin{array}{cccc} 1 & 2 & \cdots & n \end{array} \\
\begin{bmatrix}
a_{11} & a_{12} & \cdots & a_{1n} \\
a_{21} & a_{22} & \cdots & a_{2n} \\
a_{31} & a_{32} & \cdots & a_{3n} \\
\vdots & \vdots & \vdots & \vdots \\
a_{m1} & a_{m2} & \cdots & a_{mn}
\end{bmatrix}
\end{array}
$$

Find Maximum And Minimum Of An Array

- To find maximum minimum of an array you just need to sort the array.
- Use **sort()** function to sort the array.
- Default sort function sort the array from start to end.
- Aftert sorting the first element of the array is the minimum and last element of the array is maximum.

Code :

```
#include<bits/stdc++.h>

using namespace std;

int main(){
int arr[10] = { 10,20,30,40,50,32,4,0,9,23 };
sort( arr, arr+10 );
cout << "Minimum is : " << arr[0] << endl;
cout << "Maximumm is : " << arr[10-1] << endl;
return 0;
}
```

Calculate How Much Element In The Array

- You can calculate how much element in the array by calculate the size of an array and divide the size with first element size.
- Think that you have an int typearray of 10 length then you the total size of array is 40 and every element size (int) 4. So the array size is 40/4 – 10.
- int arr[] = {1,2,34,2,5,6,65,3,5,6,34};
  int len = sizeof(arr)/sizeof(arr[0]);

Code:

```
#include<bits/stdc++.h>

using namespace std;

int main(){
int arr[10] = { 10,20,30,40,50,32,4,0,9,23 };
int n = sizeof(arr)/sizeof(arr[0]);
cout << "Total element in the array is : "<< n <<
endl;
return 0;
}
```

HOMEWORK - 1

- **Find the maximum and minimum element of the given array. (Easy)**
  arr[] =
  {3,23,4,34,34,3,8,2,23,2,3,1,3,4,1,4,2,3,4,1,4,1,4,2,3,4,134,234,14,23,4,123,42,42,42,241,24,124,42,42,
  5,5,645,64,64,6,756,756,75,8,8,9,79,98,9}
  - **Hint** : count the size of the array. Sort the array. Then find the maximum and minimum.

Search In Array

```
#include<bits/stdc++.h>

using namespace std;

int main(){
int n;
cout << "Enter length of the array : ";
cin >> n;
int arr[n];
for( int i =0; i < n ; i++ ) {
cin >> arr[i];
}
cout << "Input the element you want to search : ";
int x;
cin >> x;
for( int i =0; i < n; i++ ) {
if( arr[i] == x ) {
cout << "Find in the array" << endl;
return 0;
}
}
cout << "Do not find in the array" << endl;
return 0;
}
```

Character Array

```
#include<bits/stdc++.h>

using namespace std;

int main(){
char ar[30];
cin >> ar;
cout << ar << endl;
return 0;
}
```

Practice Problem

- https://toph.co/p/copycat
- https://toph.co/p/add-them-up
- https://toph.co/p/formatted-numbers
- https://toph.co/p/clock-math

Homework 2

Input :
First take an input n (size of the array). Then take input the whole array. Now take the input x.
Output :
Print the sum of the array element from 0 to x.